

(34)

Business BI means using data to help business understand what is happening and make better decisions

BI tools

- ① - Microsoft Power BI - It makes it easy to find complex data easy to understand
- ② Tableau - Help users explore data visually and make it easier to find patterns.

(35)

Artificial Intelligence especially generative AI I found it interesting and surprising that AI can create text, images even code that looks like a human. It shows how powerful technology has become and how we can use it in education, business and everyday life

(36)

Python and coding, I use Youtube tutorials and a lot of TikTok short format videos

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Data and AI will help by making work faster and easier in the future, improving decision making and making everyday life personalized and convenient

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Bias AI

means that AI system give unfair or unbalanced results because it has learned from incomplete or unequal data

When AI is bias it means it makes decision that favors one group over another

It learns incorrect data

Examples

- AI was trained mostly on resumes from men.
- It will always favors male candidate

33 - Data engineers - build the data systems

- Data Analysts - Study and explain data
- Data Scientist - Use data to predict future outcomes
- AI Engineers - Build intelligent systems that learn and make decisions

Role	What they do	Key Skills	Common Tools
Data Engineer	Build and maintain systems collect and store data	Programming, Database, Data Pipeline	SQL, Python, Hadoop, Spark, AWS
Data Analyst	Analyzes data and create report to explain what happened	Data analysis, Statistics, Visualization	Excel, SQL, Python, Power BI, Tableau
Data Scientist	Uses data to find patterns predict future	Mechiner learning, Statistics	
AI Engineer	Build AI System that can learn and make decision	Deep learning, AI math, Programming	Python, PyTorch, C++, MATLAB

- ① Numpy - helps perform fast mathematical calculations
- ② pandas - Great for cleaning and exploring data
- ③ Matplotlib - help visualize data clearly
- ④ Scikit-learn - helps build AI that can learn from data and makes predictions

②9 Data Privacy, means protecting personal information so that it is not misused, stolen, or shared without permission
Why is data privacy is important

- It protects people from identity theft and fraud
- It keeps personal information secured
- Ensures companies use data responsibly
- It builds trust between users and organisations

GDPR - is a law created to protect people's personal data and privacy

- Countries like European Union and European Economic Area (EEA)

③ Data Ethics is about using AI in a fair, honest and responsible way and does not harm anyone.

A company collect personal information (like phone number, ID number) without telling the customer or asking for a permission

- It violate privacy
- People do not know how their data being used
- It lead to misuse of personal information

③ Data Governance

data governance is like a set rules and responsibility that control how data is collected, stored, protected and used in company

① Data Quality = Makes ^{sure} the data is clean and accurate

② Data Security = Clear keeps data safe and private

③ Data Policies and rules = Clear rules for how data should be handled

④ Data Ownership = Assign people to take responsibility for data

PaaS - { Platform as a Service }

This gives you a platform where you build, test, and deploy applications without worrying about managing the underlying hardware or operation system

SaaS - { Software as a Service }

This gives you ready made software that you can use directly over the internet without installing anything

(26) What is a Version Control System

It save history system for your work, It helps track changes made to files over time

Git: is the most popular version control system used today. It help developers and data analysts save different versions of their work and collaborate with others easily

- It tracks changes in files
- Allows team work without overwriting other work.
- Let you go back to earlier version if needed

(27) Jupyter Notebook

It is like a digital notebook where you can write code, see results immediately and explain what you are doing step by step

Jupyter notebook is popular because it is very easy for learning, testing and analyzing data.

- You can ~~code~~ code step by step and see immediately result.
- You can mix code, text, and graphs in one place

(28) Python - is a language that helps people to write code to solve problem, Analyze data, build apps and create AI system

(23) Prompt engineering - it is a ^{question} ~~tool~~ you give to an AI tool

• AI tools such as Chat GPT and Claude depend on the instructions users provides

• Poorly written Prompt

"Help me me with chicken recipe"

Improved Prompt

"I need a Roasted chicken recipe"

(24) Map Functions

is a tool in programming that is used to apply the same operation to every item in a list or collection of data
What does it do in data programming and data processing

- (i) it goes through each element in a list one by one
- (ii) Performs a function (like adding, multiplying)
- (iii) It creates a new list with results
- (iv) The original data usually stay unchanged

<1> Python

```
numbers = [1,2,3,4]
```

```
# use map to double each number
```

```
Result = list(map(lambda x: x * 2, numbers))
```

```
Print(Result)
```

```
output [2,4,6,8]
```

(25) Cloud Computing - is a way of using computer storage and software over the internet instead of owning and managing them on your own computer

IaaS (Infrastructure as a Service)

gives you basic computer resources like servers, storage and networking over the internet

Get

Used to retrieve or Read information From a server.

Post Put

Use to update or replace existing data

Post

used to send new data to a Server

Delete

Use to remove data From a Server

② JSON

is a simple format used to Store and exchange data between computers applications and API's

Why is it commonly used with API and exchange

- Easy to Read and Write
- Lightweight and simple
- Fast For computers to process

JSON object

```
{  
  "Name": "Nokwazi",  
  "age": 33,  
  "city": "Mofolo" }  
}
```

Name - Stores a persons Name

age - Stores the age.

City - store the location.

Data is written inside curly brackets

information is stored as "key value pair"

key and text value are written inside quotation marks

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Computer vision is a type of Artificial Intelligence that helps and machines understand and analyze image and videos, similar to humans use their eye to see and recognize things

Two Real world used cases of computer vision.

- Face Recognition on Smart Phones
- Self driving Cars

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An API act like messenger between two systems, It takes a Request FROM one programme Send it to another programme and then returns the response

- Weather Apps
- Online Payments
- Login with Google and Facebook

(20)

API key proves that a Program has permission to use a Service OR access certain data

- How it stored ~~and~~ protected and expose

- use your account without permission {exposed}
- Stored it in secure file {Protected}
- cause unexpected charges or cost {exposed}
- Never Share it publicly {Protected}

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A REST API is a type of API that allows applications to communication with each other over using rules and Standard Web.

REST Stand for
Representational State Transfare

Example in everyday life

- Voice Assistance
- Translation Apps
- Chatbots and Customer Supports

(15) is a type of Artificial Intelligence (AI) that is trained to understand and generate human language.

Well known LLMs

- Chat GBT
- Google Gemini

(16) Generative AI is technology that can create content similar to what human make, It also generate and or produce New things base on what it has learned

Generative AI is trained using huge amount of data such as books, pictures, videos or Website.

Chat GBT - Used for answering questions, writing essays, Creative writing

DALL-E - used to create images and Artwork from text descriptions given by users

Google Gemini -

Use to generating text, answering questions, Summarizing information, helping users across Google products and Service.

(17) It is training dataset is like study material that helps the AI learn patterns and make decisions. Because good data leads good AI it is always good to have quality of training data.

If an AI is trained wrong/with incorrect information It may give wrong answers

12) Machine learning

is a branch of artificial intelligence where computers learn from data and improve automatically without being explicitly programmed for every task

- Supervised

The machine learns by inputting data and correct output labels by using questions and correct answers

- Unsupervised

The machine receives data but without answers and it must find answer on its own.

- Reinforcement Learning

The machine learns from errors, it is a robot learning

13) Deep Learning

- It teaches computers to learn patterns from huge amounts of data especially for tasks like
- Recognizing Images
- Understanding Speech
- Translating Language
- Driving Cars

How does it relate

AI : It is the broad field

ML : is a subset of AI

Deep Learning : is a subset ML.

14) Natural language Processing - enable computers and digital devices to recognise, understand and generate text speech by combining computational linguistics the rule based modeling of human language together with statistical modeling

- (10) It is a Method for to organize and store data
So computers can find and use it efficiently
- Different formats are useful for different situations

CSV (Comma-Separated Values)

data is stored as plain text with values separated by comma
eg Name, Age, City

JSON (JavaScript Object Notation)

It stores data in a structured and flexible way using key-value pairs

eg "Name" Jason
"Age" 33
"City" Soweto

Parquet

It is modern format designed for big data analytics
Unlike CSV it stores data by column instead of Row.
eg Thabo, 25, Soweto
Tumi, 30, Mlazi

Avro

is a data format designed for fast data exchange and streaming system

(11) Artificial Intelligence (AI)

It is the ability of computers or machine to perform tasks that normally require human intelligence

- Narrow AI : is designed to perform one specific task very well

eg Siri can answer questions, set alarms but can not drive a car

- General AI : is a theoretical type of AI that can think, learn, understand, reason just like human being across different tasks

⑧ DBMS is software that helps users to store data, organize data, retrieve data, update data and manage data securely and efficiently. It act like a smart digital system.

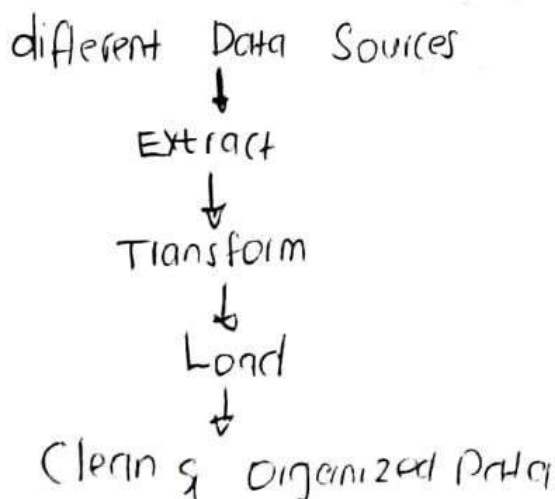
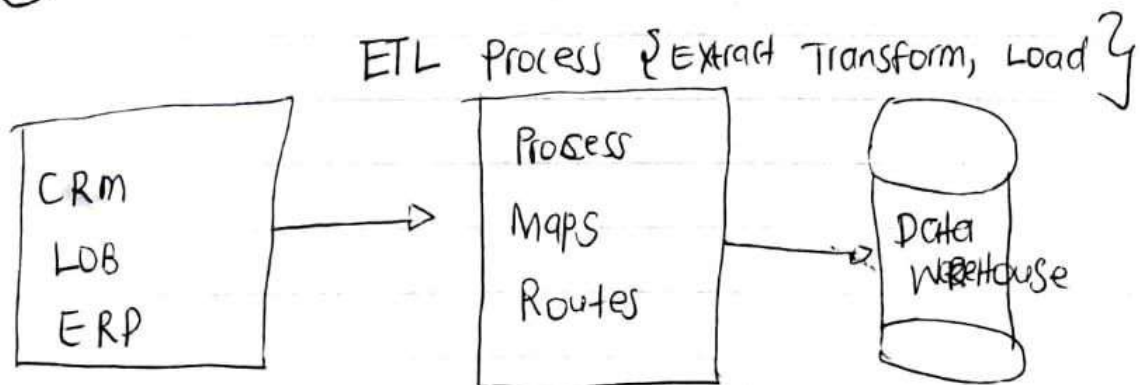
My SQL : Stores Tables, Rows and Columns

Postgre SQL : It is a powerful open source DBMS known for advanced features and reliability

Oracle Database - it handles very large enterprisos systems, very powerful, high performance and advance security

MongoDB - instead of table it stores data in a flexible document like format

⑨ ETL helps business turn raw data into useful information



- Structural data - It is stored in a table, It is consistent
- Unstructured data - It does not have consistence format and it is unorganised.
- Semi Structured data - It does not reside from a traditional relational data base.

EXAMPLES

STRUCTURED	Unstructured	Semi Structured
• Numbers and date • Relational database • Less	• Data is fully non organized • Difficult, but achievable transaction management and data concurrency	• Data is more Flexible than Structured data • Data is organized to some extent.

⑥ ~~Data~~ Modeling is the process of creating a visual Representation of either a whole information system or part of it to communicate connections between data points and structures

types of data modeling

- Conceptual data models
- Logical data models

⑦ Data mining it is the process of finding ~~these~~ ^{the} hidden patterns, trends and useful information from a large amount of data

- Digging through the data to find valuable knowledge

Example:

Business collect huge amount of data everyday from

- * Online Shopping
- * Mobile Apps

Data Analytics & Data Science

Research Assignment

① It is a collection of organised information, it helps with efficiency, updated data/consistency and security
 Example of where the databases are used

- Fleet management companies
- Health insurance information

② What is a Data Warehouse

- It is a centralized system designed for storing, managing and analyzing large volumes of data from multiple sources
- Companies use data warehouse, to make decisions in hiring, market+sales, pricing and customer support

③ What is data Lakes

It is a vast storage repository that stores a huge amount of raw data in its native format until it is needed
 data lake stores, ~~structured~~, structured and unstructured data like social media posts whilst data warehouse stores information or data in tables and hierarchy.

④ What is a data Mart

- It is a subgroup of a data warehouse that is designed to support the business intelligence and reporting needs of a specific group of users within an organisation
 It is used by • It
- It is a smaller version of data warehouse, used by a single department or function, focus on business function for a specific user group in an organisation

⑤ Explain the difference between Structured, Semi Structured and Unstructured data.